AIM

# TA.120 PLATFORM AND NETWORK ARCHITECTURE

# <Company Long Name>

# <Subject>

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# Contents

# Introduction

## Purpose

The purpose of the **Platform and Network Architecture** document is to describe the physical platform and network configuration to support the final Platform and Network architecture. You map the physical application and database server architecture onto specific computing platforms included in the scope of the <Project Name> Application and Technical Architecture process.

## Scope

The <Project Name> platform and network architecture work covers the following topics:

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# Platform and Network Standards

## Platform Standards

This section describes the standards that the <Project Name> project platforms should meet.

These standards were arrived at by:

The platforms standards cover the following components and hardware types:

- hardware platform servers
- Internet/web servers
- data storage media
- networking devices
- networking protocols

The standards for platform servers are summarized in the following table:

| Platform Server Class | Hardware Standard | Vendor | Operating System | Platform Resources | Comments |
|---|---|---|---|---|---|
| Database Server | | | | | |
| Concurrent Management | | | | | |
| Administration | | | | | |
| Web | | | | | |
| Forms | | | | | |
| Printing | | | | | |

## Network Standards

This section describes the standards that the <Project Name> project networks should conform to.

These standards were arrived at by:

The network standards cover the following network components:

- Wiring
- Protocols
- Routers
- Bridges
- Modems

## Wiring

The standards for network wiring are summarized in the following table:

| Application/ Circuit Type | Network Standard | Network Speed | Cable Type | Comments |
|---|---|---|---|---|
| LAN | Ethernet | 10Mb/sec, 100 Mb/sec, Gigabit | UTP, STP, Coaxial | 10BaseT, 100BaseT, 10Base2, 10Base5, Gigabit Ethernet |
| LAN | Token Ring | 4Mb/sec, 16Mb/sec | UTP, STP | IBM "standard" |
| LAN | Arcnet | 2.5Mb/sec | UTP | Non-IEEE standard |
| LAN | LocalTalk | 232Kb/sec | UTP | Apple Computer's Peer-to-Peer networking "standard" running Apple's AppleTalk protocol |
| LAN | EtherTalk | 10Mb/sec, 100Mb/sec, Gigabit | UTP | Apple Computer's Implementation of AppleTalk over Ethernet. EtherTalk can support both AppleTalk as well as standard Ethernet hosted protocols |
| LAN | FireWire | 400Mb/sec | FireWire Cable | Supported by Apple, and other hardware manufactures. Often used in high-bandwidth multimedia applications. Hot-pluggable. Greater than a 600 device limit. Self-Configuring. |
| LAN, Device Bus | Universal Serial Bus | 12/Mb/sec | UTP/USB | Used for device bus. May also be used for rudimentary file transfer. Hot-pluggable. 127 device limit. |
| LAN | FDDI/CDDI | 100Mb/sec | Fiber Optic, Copper | Common "Backbone" use |
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**Note:** Kb/sec = kilobits per second, Mb/sec = megabits per second, UTP = Unshielded Twisted Pair, STP = Shielded Twisted Pair, XXBaseT = Twisted Pair cabling solution, 10Base2 = Thin Coaxial cabling solution, 10Base 5 = Thick Coaxial cabling solution

## LAN Transport and Protocols

The standards for LAN transport and protocols are summarized in the following table:

| Network/Host Operating System/Environment | "Native" Network/Transport Protocols | Application/Circuit Type | Comments |
|---|---|---|---|
| Novell NetWare | IPX/SPX | | |
| IBM LAN Server | NETBEUI | | |

| Network/Host Operating System/Environment | "Native" Network/Transport Protocols | Application/Circuit Type | Comments |
|---|---|---|---|
| Windows/NT | NETBEUI, TCP/IP, IPX/SPX | | |
| DEC Pathworks | DECnet, NETBEUI | | |
| IBM Mainframe | SNA | | |
| Apple MacOS, MacOS-X Server and Client Edition | AppleTalk, IPX/.SPX, TCP/IP, NETBEUI (with MacOS-X Server) | | |
| | | | |
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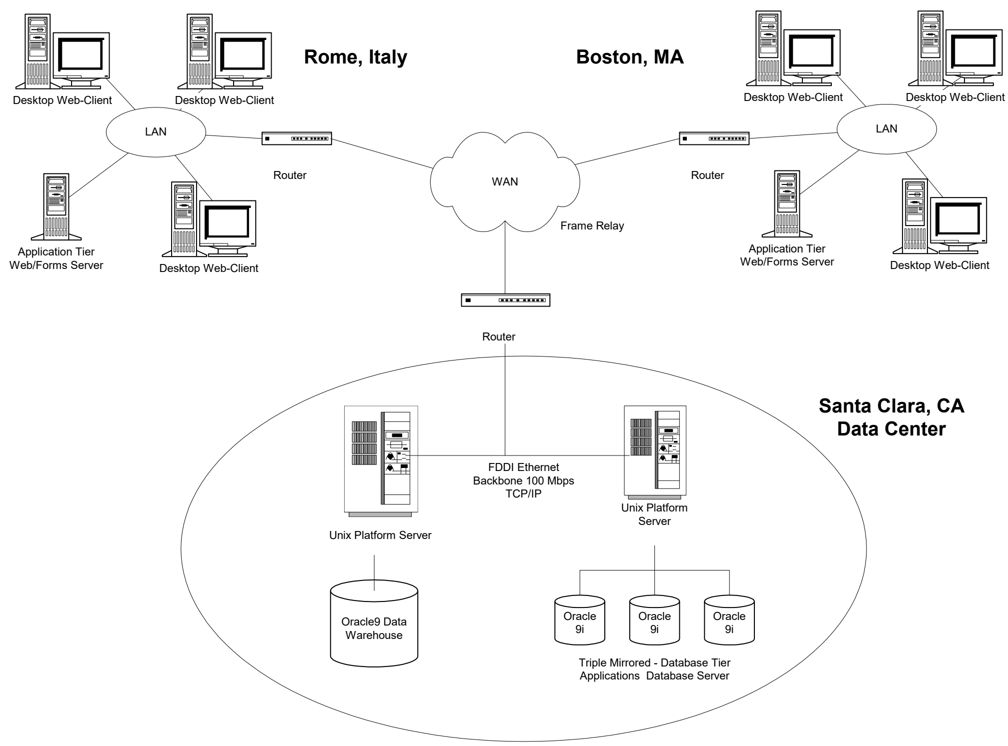
### WAN Transport and Protocols

The standards for WAN transport and protocols are summarized in the following table:

| WAN protocol standard | Bandwidth | Application/Circuit Type | Comments |
|---|---|---|---|
| Frame Relay | T1- 1.54 Mbps | Point-to-Point | |
| ISDN | 128 Kbps | | |
| ATM | T1-T3 | | |
| PPP | | | On-Demand Dial or fixed |
| | | | |

# Enterprise Platform and Network Architecture

## Technical Configuration




### Client Desktop Environment

Clients will be primarily PCs running Microsoft Windows located at the work sites.

## Platform Deployment

### Servers

The main database servers will be located at: <Location>.

The configuration of each server is detailed below.

| Server | Model | Number of CPUs | Disk Space | Disk Controller | Total RAM | Backup Media |
|---|---|---|---|---|---|---|
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## Network Connections

The Enterprise Network Architecture of <Company Short Name>'s network is of the <Network Type> type.

| Site 1 | Site 2 | Network Type | Protocol | Band width | Latency | Network Hops | Comments |
|---|---|---|---|---|---|---|---|
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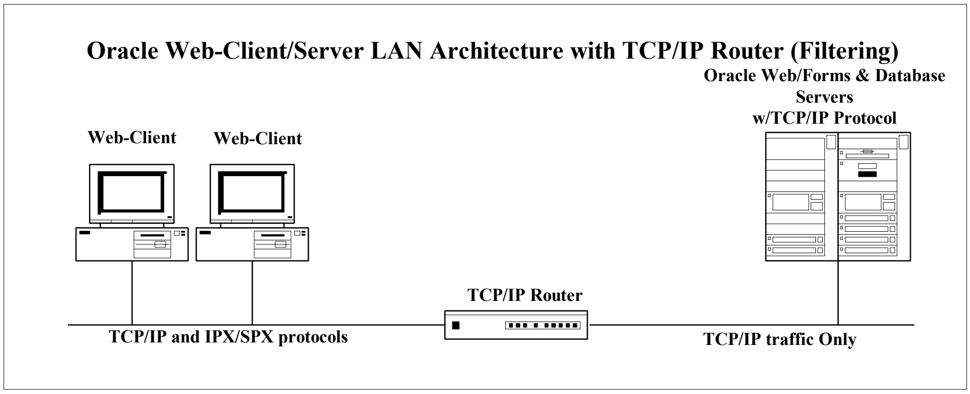
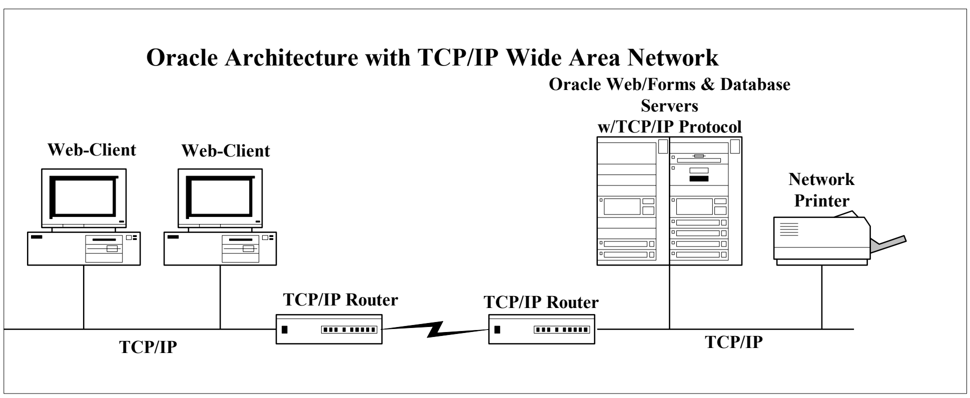
# Platform and Network Architecture for <Data Center/Hosting Facility Name>

## Technical Configuration




## Database/Application Tier

### Server Mapping

This section describes the servers that support each application.

| Application Type | Application Environment | Platform Type | Platform DNS Name | O/S Version | Facility |
|---|---|---|---|---|---|
| Database | PROD | Sun E-1250 | PROD1 | Solaris 7.5 | Corporate |
| Concurrent | PROD | Sun E-1250 | PROD1 | Solaris 7.5 | Corporate |
| Administration | PROD | Sun E-1250 | PROD1 | Solaris 7.5 | Corporate |
| Web | PROD | Compaq Proliant 2300 | PRODW1 | NT3000 | Corporate |
| Web | PROD | Compaq Proliant 2300 | PRODW2 | NT3000 | Corporate |
| Web | PROD | Compaq Proliant 2300 | PRODW3 | NT3000 | Corporate |
| Forms | PROD | Compaq Proliant 2300 | PRODW1 | NT3000 | Corporate |

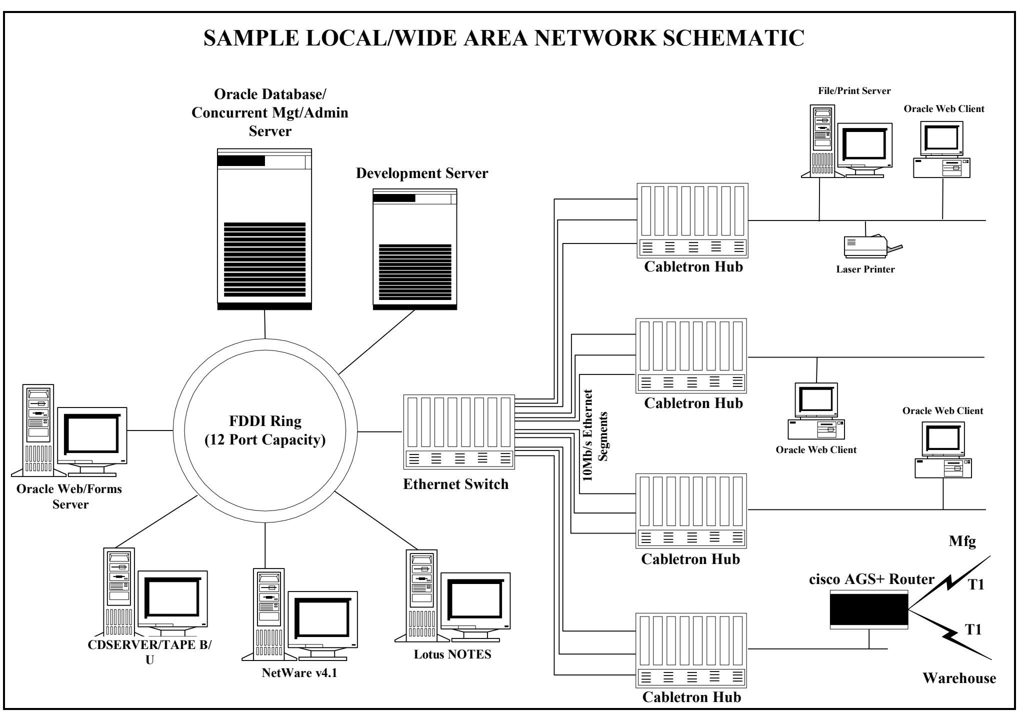
| Application Type | Application Environment | Platform Type | Platform DNS Name | O/S Version | Facility |
|---|---|---|---|---|---|
| Forms | PROD | Compaq Proliant 2300 | PRODW2 | NT3000 | Corporate |
| Forms | PROD | Compaq Proliant 2300 | PRODW3 | NT3000 | Corporate |
| | | | | | |
| | | | | | |

### Platform Server Configuration

The configuration of each platform is detailed below.

| Server | Model | No. of CPUs | Disk Space | Disk Controller | Total RAM | Backup Media |
|---|---|---|---|---|---|---|
| PROD1 | Sun E-1250 | 8 | 32GB | 4-Ch/SCSI | 4GB | 12GB DLT |
| PRODW1 | Compaq Proliant | 4 | 12GB | 2-Ch/SCSI | 2GB | 12GB DLT |
| PRODW2 | Compaq Proliant | 4 | 12GB | 2-Ch/SCSI | 2GB | 12GB DLT |
| PRODW3 | Compaq Proliant | 4 | 12GB | 2-Ch/SCSI | 2GB | 12GB DLT |
| | | | | | | |
| | | | | | | |

## Data Center Network Configuration




### Network Specifications

| Site 1 | Site 2 | Network Type | Protocol | Band width | Latency | Network Hops | Comments |
|---|---|---|---|---|---|---|---|
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# Client Desktop Tier for <Deployment Site Name>

## Client Desktop Environment

Clients will be primarily PCs running Microsoft Windows located at the work sites. The diagram above illustrates the locations of all components. These systems will be running Netscape Communicator version 9.5 and will be configured with 48MB of RAM and 35 MB of available disk space. They will be connected to the corporate LAN/WAN at 10MBS.

## Printer Deployment

The printers that will be used for the production system report printing are of the types:

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The specifications of the printers are:

The printer deployment strategy is:

## Peripheral Devices

The types of peripheral devices that are part of <Company Short Name>'s <Project Name> technical architecture are:

- Modems
- 
- 

The impact of the peripheral devices on the technical architecture is:

# Open and Closed Issues for this Deliverable

## Open Issues

| ID | Issue | Resolution | Responsibility | Target Date | Impact Date |
|---|---|---|---|---|---|
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## Closed Issues

| ID | Issue | Resolution | Responsibility | Target Date | Impact Date |
|---|---|---|---|---|---|
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